ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

BIOGRAPHY OF SUNDAR PICHAI

THE ARCHITECT OF EMPOWERMENT & INTEGRATION
An Engineering & Leadership Biography

Full Name: Pichai Sundararajan (Sundar Pichai)
Born: July 12, 1972 | **Birthplace:** Madurai, Tamil Nadu, India
Profession: Technology Executive
Current Role: CEO of Alphabet Inc. and Google

Sundar Pichai (Pichai Sundararajan) is the Chief Executive Officer (CEO) of Alphabet Inc. and its subsidiary Google. Rising from a humble background in Chennai, India, his career trajectory stands as a definitive blueprint for modern software engineering, product management, and technological integration. Pichai's legacy is defined not just by creation, but by integration—taking complex, core technologies and embedding them seamlessly into global ecosystems. For aspiring AI and Model Integration Specialists, his professional methodology offers a masterclass in infrastructure design, system orchestration, and collaborative engineering.

1. Early Life and Academic Foundations

He pursued his foundational engineering degree at the Indian Institute of Technology (IIT) Kharagpur, where he earned a B.Tech in Metallurgical Engineering. Recognizing his exceptional grasp of material structures and analytical thinking, his professors recommended he pursue advanced studies in the United States. He was awarded a scholarship to Stanford University, where he earned an M.S. in Materials Science and Engineering, followed by an MBA from the Wharton School of the University of Pennsylvania, where he was named a Siebel Scholar and a Palmer Scholar.

2. The Google Meteoric Rise: An Integration Masterclass

Pichai joined Google in 2004 as a Product Manager. His initial responsibilities centered around the search toolbar, a seemingly minor project that would eventually solidify Google's dominance on the internet.

A. The Google Toolbar: Connecting Core Technology

In 2004, Google's core asset—its search engine—replicated its utility primarily through third-party web browsers like Microsoft Internet Explorer. Pichai led the team responsible for the Google Toolbar. This was fundamentally an *integration project*. It safely embedded Google's search capabilities directly into competitor ecosystems. When Microsoft threatened to change its default search engine to Bing, Pichai's integration strategy saved Google's user base by allowing seamless access to Google Search without requiring users to navigate away from their active browser workflow.

B. Google Chrome: Building the Infrastructure Layer

Recognizing that web browsers were highly inefficient at running complex web applications, Pichai proposed building a proprietary browser. Despite initial skepticism from executive leadership regarding the necessity of entering a 'browser war,' Pichai persisted. In 2008, Google launched Chrome. Under his guidance, Chrome was built with an advanced, lightning-fast JavaScript execution engine (V8). Rather than just a viewing portal, Chrome was designed as an infrastructure layer capable of executing cloud applications smoothly. This mirrors the modern challenge of a Model Integration Specialist: ensuring that underlying pipelines are fast enough to run complex workloads (like LLMs) seamlessly in real time.

C. Android and Ecosystem Orchestration

In 2013, Pichai took over the management of Android following Andy Rubin's departure. Android was a powerful operating system, but it required systemic harmonization across thousands of different hardware manufacturers, carriers, and app developers. Pichai successfully unified the ecosystem, ensuring consistent developer APIs, secure cloud connections, and open-source accessibility. This project demonstrated his elite capacity for ecosystem orchestration—bringing disparate software components together into a single, cohesive framework.

3. The Leap to CEO: Steering the 'AI-First' Revolution

In August 2015, during Google's corporate restructuring into Alphabet Inc., Sundar Pichai was appointed CEO of Google. In December 2019, he assumed the role of CEO of Alphabet Inc. as well. Upon taking the helm, Pichai made a historic, predictive pivot. He declared that Google would transition from a 'Mobile-First' company to an **'AI-First'** company. Under his direction, Google integrated machine learning and artificial intelligence into its entire product suite:

- **Google Search:** Enhanced with deep learning models (such as BERT, MUM, and Gemini) to understand conversational user intent.
- **Google Cloud:** Empowering enterprise developers to deploy, integrate, and manage complex machine learning architectures via Vertex AI.
- **Google Gemini:** Deploying unified, multimodal AI models across consumer applications like Android, Maps, and Workspace.

Pichai's leadership style is characterized by deep technical empathy, collaboration, and diplomatic execution. He is renowned for navigating highly technical corporate environments by prioritizing consensus, encouraging cross-departmental collaboration, and empowering engineering teams to solve complex infrastructure hurdles.

4. Key Strategic Takeaways for the Modern AI Specialist

1. **System Interoperability is King:** A core technology—no matter how powerful—is only as valuable as its connection to the end user. Just as Pichai connected search via the Toolbar, an AI Integration Specialist connects complex neural networks to functional business applications.
2. **Optimize the Infrastructure:** High latency kills user adoption. Building Google Chrome taught the world that client-side execution environments must be ruthlessly optimized. Modern AI engineering requires the exact same focus when connecting models to high-performance Vector Databases and low-latency API architectures.

CAREER PROFILE

MODEL INTEGRATION SPECIALIST (AI ENGINEER)

Name: Harshitha V
College: St. Joseph's College of Engineering, Chennai
Department: B.E Computer Science and Engineering (CSE) | **Year:** I
Email: harshithavenugopal3107@gmail.com | **Location:** Chennai

Career Objective

To become a skilled and innovative Model Integration and AI Engineer by building a powerful foundation in computer science, software engineering, and cognitive computing. Aiming to leverage strong backend, database, and API development skills to securely connect, orchestrate, and deploy production-ready artificial intelligence models within enterprise software architectures by 2030.

My 4-Year Career Roadmap

YEAR 1 - FOUNDATION: Learn programming fundamentals and algorithmic problem-solving; develop core skills in Python, C, HTML, CSS, and basic JavaScript; master version control using Git and GitHub; build small foundational automation scripts and data-driven projects.

YEAR 2 - SKILL DEVELOPMENT: Master Data Structures and Algorithms (DSA) and Object-Oriented Programming (OOP); develop deep knowledge of SQL, Relational DBMS (PostgreSQL), and backend web development; begin learning AI application frameworks like LangChain and LlamaIndex; build custom API wrappers and maintain an active GitHub portfolio.

YEAR 3 - SPECIALIZATION & INDUSTRY EXPERIENCE: Strengthen advanced programming, Operating Systems, and Computer Networks; develop full-stack applications with robust REST/gRPC API architectures; integrate Vector Databases (Pinecone, Milvus) and explore cloud fundamentals (AWS/GCP AI tools); secure internships focused on AI engineering, RAG pipelines, or backend infrastructure.

YEAR 4 - CAREER & PLACEMENTS: Complete an industry-oriented final-year project featuring a deployed, optimized machine learning or generative AI workflow; prepare for advanced technical interviews covering system design, cloud MLOps orchestration, and core CS fundamentals; refine communication and technical presentation skills to launch a definitive career as a Model Integration Specialist.

Key Skills

- **Programming Languages:** Python, Java, C, C++, JavaScript
- **AI & Integration (Target):** LangChain, LlamaIndex, REST/GraphQL/gRPC APIs, Prompt Orchestration
- **Database & Vector Data:** SQL, MySQL, PostgreSQL, Vector Databases (Chroma, Pinecone, Milvus)
- **Tools & Infrastructure:** Git, GitHub, VS Code, Docker, Cloud AI Gateways (AWS, GCP)

Future-Proof Career Opportunities (2030)

Core Focus: Model Integration Specialist | AI Application Engineer | Backend MLOps Engineer

Secondary Paths: Data Infrastructure Engineer | Cloud Solutions Architect | Full-Stack AI Developer

Long-Term Path: CSE Student → AI/Backend Intern → Model Integration Specialist → AI Systems Engineer → Principal AI Architect → CTO

My Vision

To continually engineer intelligent, highly scalable software infrastructures that seamlessly bridge human enterprise workflows with advanced artificial intelligence, creating meaningful digital ecosystems that solve complex, real-world industry problems by 2030.

SWOC ANALYSIS

ASPIRING MODEL INTEGRATION SPECIALIST

STRENGTHS

• **Strong interest** in technology, programming, and system integration.

• **Good problem-solving** and analytical thinking skills.

• **Willingness** to learn APIs, integration tools, platforms, and new technologies.

• **Ability to work in teams** and communicate effectively with different teams.

• **Adaptability** to changing technologies and evolving software environments.

WEAKNESSES

• **Limited industry experience** as a student.

• **Need to improve** advanced programming, API, and integration skills.

• **Limited real-world project exposure** to large-scale integration systems.

• **Time management** between academics and technical skill development can be challenging.

• **Need to build confidence** in technical presentations and professional discussions.

OPPORTUNITIES

• **Growing demand** for specialists who connect applications, services, APIs, and AI models.

• **Internship opportunities** in software integration, cloud, automation, and AI-related roles.

• **Career paths** in API integration, cloud platforms, AI/ML model integration, and enterprise systems.

• **Availability** of online courses, developer tools, documentation, and hands-on integration projects.

• **Possibility** of working with global teams and contributing to technology-driven products.

CHALLENGES

• **High competition** for technology and integration-related roles.

• **Rapid changes** in APIs, cloud platforms, AI models, and integration technologies require continuous learning.

• **Balancing** academics, projects, certifications, and placement preparation.

• **Keeping skills updated** with new tools, frameworks, security practices, and industry standards.

• **Solving** complex system-compatibility and integration problems under deadlines.

CONCLUSION

This SWOC analysis helps me understand my current abilities, areas for improvement, future opportunities, and possible challenges in achieving my goal of becoming a successful **Model Integration Specialist**. By continuously learning programming, APIs, integration platforms, cloud technologies, and practical problem-solving, I can strengthen my skills and build a successful career in model integration.

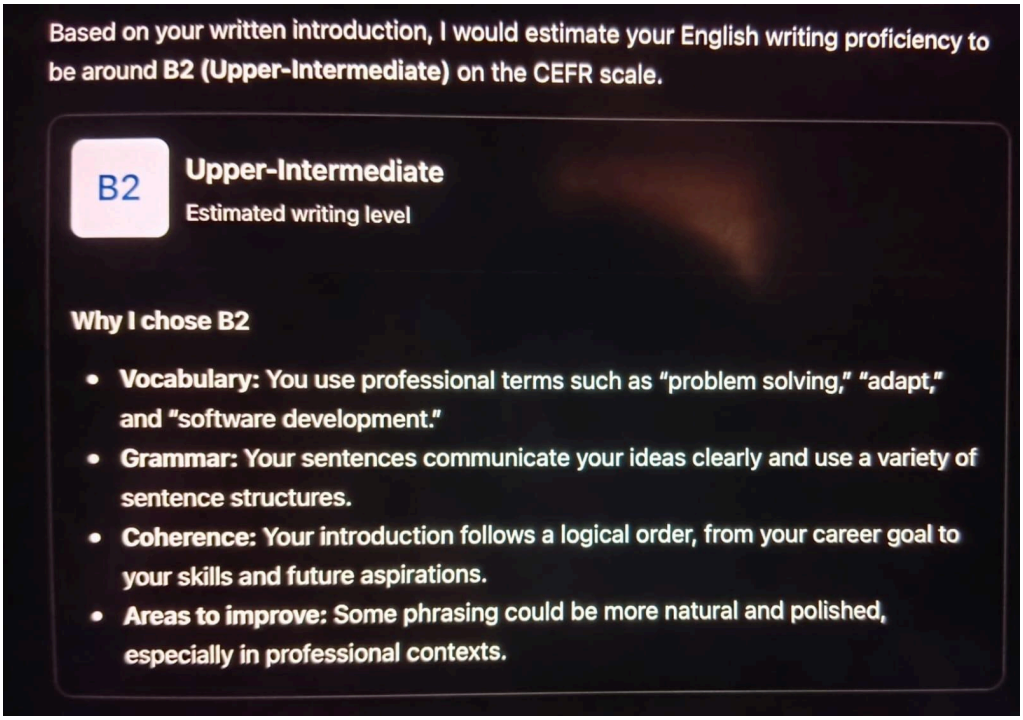
My Experience with Write & Improve

My experience with Write & Improve was simple, useful, and informative. I did not find the activity very challenging, and I was able to express my ideas in written English comfortably. It gave me an opportunity to practise writing and understand how effectively I could communicate my thoughts.

The feedback provided after writing helped me identify areas that could be improved. In particular, it made me pay more attention to grammar, vocabulary, sentence structure, spelling, punctuation, and clarity of ideas. I also understood the importance of reviewing and correcting my writing instead of submitting the first version.

Another useful part of the activity was that it encouraged me to think about how my sentences could be made clearer and more effective. It helped me become more aware of repeated mistakes and showed me how practice can gradually improve my writing skills.

Overall, Write & Improve was a valuable learning experience. It helped me evaluate my written English, understand my strengths and areas for improvement, and motivated me to practise writing regularly to become a better and more confident English communicator.


Based on your written introduction, I would estimate your English writing proficiency to be around B2 (Upper-Intermediate) on the CEFR scale.
B2
Upper-Intermediate
Estimated writing level
Why I chose B2
Vocabulary: You use professional terms such as "problem solving," "adapt," and "software development."
Grammar: Your sentences communicate your ideas clearly and use a variety of sentence structures.
Coherence: Your introduction follows a logical order, from your career goal to your skills and future aspirations.
Areas to improve: Some phrasing could be more natural and polished, especially in professional contexts.

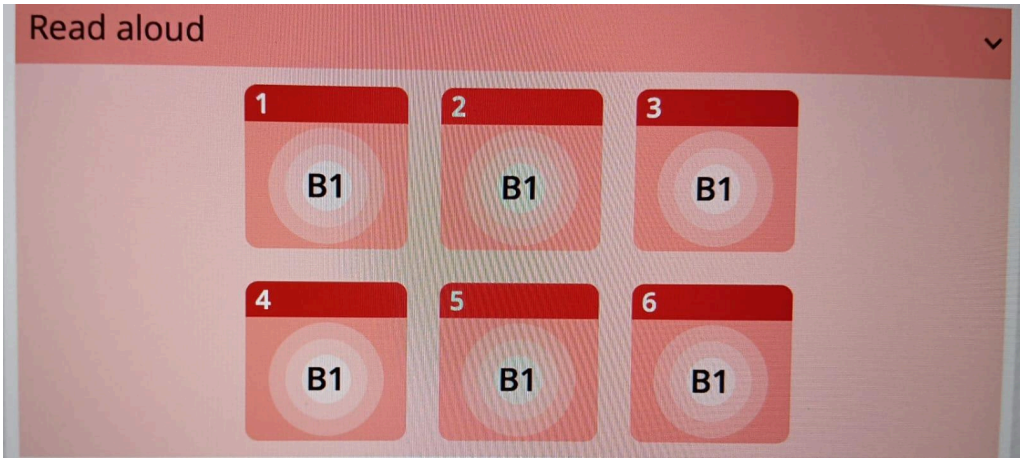
My Experience with Speak & Improve

My experience with Speak & Improve was interesting and helpful. I did not find the activity very challenging, as I was able to understand the questions and express my thoughts comfortably. It provided me with an opportunity to practise speaking in English without feeling too pressured.

I received a B1 level according to the CEFR. The result helped me understand my current level of English communication. Through this activity, I was able to reflect on my fluency, pronunciation, vocabulary, grammar, and confidence while speaking.

One of the useful aspects of Speak & Improve was that it gave me an idea of the areas where I need further practice. It also encouraged me to speak more naturally and organise my thoughts before expressing them. I realised that regular speaking practice can help me become more confident and fluent.

Overall, Speak & Improve was a good experience for assessing my speaking ability. My B1 CEFR result motivated me to continue practising and work towards achieving a higher level of English proficiency in the future.


Read aloud
1
B1
2
B1
3
B1
4
B1
5
B1
6
B1

My Experience with Enhancing Soft Skills and Personality

NPTEL course by Prof. T. Ravichandran, IIT Kanpur · 8 weeks

Introduction

This course is about the “people side” of success: how we think, manage our time, deal with others, and take care of ourselves. Technical skills help us get a job, but soft skills help us keep it and grow in it. The course ran for eight weeks, with short video lectures and a weekly assignment. Below is a simple week-wise summary of what I learned.

Week 1: Mindset

The course began with a quick revision of the earlier “Developing Soft Skills and Personality” course, followed by lessons on mindset. I learned the difference between a fixed mindset (“I can't do this”) and a growth mindset (“I can learn this”). The main takeaway was that abilities are not fixed. With effort, practice and a willingness to learn from mistakes, anyone can improve.

Week 2: Time and Procrastination

This week was about how we see and use time. I learned why people delay tasks and how procrastination slowly hurts work and confidence. The lectures gave practical ways to plan the day, set priorities and start tasks early. One interesting lesson was that saying “yes” to everyone just to keep them happy wastes our own time and energy.

Week 3: People, Anger and Positive Thinking

We looked at the different types of people we meet and how to deal with each of them. I learned how to say “no” politely but firmly without hurting relationships. The lectures on controlling anger were very useful, because they showed how a calm response solves problems better than a quick reaction. The week ended with the power of positive thinking and how our attitude shapes our results.

Week 4: Being Likeable

This week asked a simple question: why do people like or dislike us? I learned that small habits, such as being rude, interrupting, or complaining too much, can push people away. In contrast, good listening, respect, a smile and genuine interest make us more likeable. The lessons on being attractive explained that real attractiveness comes from confidence, behaviour and personality, not only looks.

Week 5: Common Errors

The whole week focused on common errors that we make in communication and daily behaviour without even noticing. Each lecture pointed out one set of mistakes and showed the better way to handle it. This week made me more careful about how I speak and write, and it showed that small corrections can make a big difference to how others see us.

Week 6: Humour and Money

I learned that humour is a strong communication tool. Used well, it reduces stress, builds friendships and makes the workplace lighter. The lectures also warned that humour should never hurt or insult anyone. The second half dealt with money and personality, where I learned the basics of managing money wisely: earning, saving and spending with discipline.

Week 7: Health and Love

This week reminded me that a good personality needs a healthy body and mind. The lectures covered the importance of regular exercise, a balanced diet and enough sleep. The second part dealt with love and relationships, and how to manage emotions and expectations in a mature way. It was a refreshing change from the usual workplace topics.

Week 8: Ethics, Etiquette, Memory and Environment

The final week covered personal and business etiquette, such as good manners, punctuality and respectful behaviour at work. I also learned simple techniques to manage the mind and improve memory. The course ended with the need to care for the environment and a quick review of all the main ideas.

Key Takeaways

- Believe that you can grow, and keep learning.
- Manage time well and avoid delaying tasks.
- Stay calm, think positively and learn to say “no” politely.
- Be respectful and good-natured, and people will like working with you.
- Look after your health, money, relationships and surroundings.

Conclusion

The course was simple, practical and easy to follow. Every week gave me ideas I could apply in real life right away, not just theory. Most of all, it helped me see that personality is not something we are born with. It is built through small, regular habits. I would recommend it to students and working professionals alike.

IDEATION: RESQMESH

An Offline-First Disaster Communication Network

1. Professional Problem Statement

During natural disasters, large public events, infrastructure failures, or other emergencies, communication networks may become unreliable or unavailable. Cellular towers can be damaged or overloaded, internet connectivity may be interrupted, and people may be unable to contact family members, volunteers, or response teams. However, even in these conditions, people still need to communicate short and important updates such as **“I am safe,” “I need help,” “This route is blocked,”** or **“Please send assistance.”**

Most commonly used messaging services depend on internet access or cellular infrastructure. When these services fail, a message may not reach its recipient at the time it is needed. This creates a communication gap for people in affected areas and makes coordination more difficult for communities and volunteers. The challenge is therefore to design a practical communication platform that can continue exchanging critical information locally, even when conventional internet or mobile network connectivity is unavailable.

Problem statement: To design an offline-first communication platform that enables nearby smartphones to exchange and relay short, critical messages during internet or cellular network outages.

2. Proposed Solution – ResQMesh

ResQMesh is proposed as an **Offline-First Disaster Communication Network**. It is designed to allow nearby participating smartphones to form a temporary peer-to-peer (P2P) communication network. Instead of depending entirely on a central server or public internet connection, devices can discover nearby peers and pass messages from one device to another when a communication path becomes available.

The system focuses on small, high-priority text messages and emergency status updates. A message is saved locally on the sender’s device and can be relayed through reachable participating devices until it reaches the intended recipient. This store-and-forward approach means that a message does not have to fail immediately just because the recipient is not directly reachable at that moment.

How ResQMesh Works

1. Create: The user writes a short message or selects an emergency status.
2. Discover: The application identifies nearby ResQMesh participating devices.
3. Queue: The message is stored locally with a unique message ID.
4. Relay: Available peers forward messages they have not already received, allowing hop-by-hop delivery.
5. Deliver: The recipient receives the message, and acknowledgements help prevent unnecessary duplicate forwarding.

RESQMESH: ADVANTAGES AND CONCLUSION

A practical concept for communication during connectivity outages

3. Advantages of ResQMesh

1. **Offline-first communication:** ResQMesh is designed to support local message exchange without requiring an active internet connection or continuous access to cellular infrastructure.
2. **Uses existing smartphones:** The concept makes use of devices that people already carry, reducing the need for dedicated communication hardware for the initial prototype.
3. **Store-and-forward delivery:** Messages can remain queued on participating devices and be forwarded when another peer becomes reachable. This can improve the chance of delivery in a changing network environment.
4. **Priority-aware messaging:** Emergency messages can be prioritised over routine updates so that urgent information receives attention within the local network.
5. **Simple and focused communication:** The proposed minimum viable product concentrates on short, useful messages such as safety confirmations, requests for help, and route alerts rather than bandwidth-heavy media.
6. **Potential for wider use:** In addition to disaster situations, the concept may be useful in remote locations, college campuses, large events, and volunteer operations where temporary local communication is valuable.
7. **Scope for synchronization and coordination:** When normal internet connectivity becomes available again, the system may synchronize queued information. Institutional features such as managed groups and an administrative dashboard could support organized deployments.

4. Practical Considerations

ResQMesh is a proposed solution and its performance must be established through development and testing. A prototype should be evaluated using a small group of devices, first with a basic local connection and then with store-and-forward messaging. Important measures include the percentage of messages delivered within a defined time, delivery delay, battery consumption, reliability as connectivity changes, and ease of use. Device range, background operation, operating-system permissions, privacy, and security also require careful validation. ResQMesh is intended as a supporting communication layer and not as a replacement for official emergency services or established emergency communication systems.

5. Conclusion

Reliable communication is essential when people face uncertain and difficult situations. Yet conventional communication channels may become unavailable precisely when timely information is most important. ResQMesh addresses this challenge by proposing a temporary, decentralised network that allows nearby smartphones to store, relay, and deliver short messages without depending on continuous internet access.

The key strengths of the concept are its offline-first approach, use of existing devices, store-and-forward message delivery, and support for prioritising emergency updates. With a carefully developed prototype and measurable testing, ResQMesh can be evaluated for use in disaster preparedness, campus safety, remote communities, and volunteer coordination. The next step is to build a small minimum viable product, test it under simulated network-outage conditions, collect user feedback, and improve its reliability and usability.

ResQMesh expresses a simple idea: when conventional networks fail, communication should still have a way to continue.

GITHUB README (HarshithaV31)

Repository Profile & Markdown View

Hi, I'm Harshitha Venugopal! ■

Aspiring Software Engineer | CSE Student (2026-2030)

I am a first-year Computer Science Engineering student passionate about problem-solving and software development. I am currently focusing on mastering Data Structures & Algorithms (DSA) and expanding my multi-language programming foundations.

■ Education

B.E. Computer Science Engineering at St. Joseph's College Of Engineering

■ Currently Learning

Data Structures & Algorithms (DSA) & Object-Oriented Programming (OOP)

■ Core Languages

Java, C++, C, Python

■■ Dynamic Skills Tracker

- Mastering Data Structures and Algorithms (DSA) & Object-Oriented Programming (OOP).
- Deepening knowledge of SQL, Relational DBMS (PostgreSQL), and backend development.
- Exploring AI application frameworks like LangChain and LlamaIndex while building custom API wrappers.

■ Specialization & Industry Experience Target

- Strengthening advanced programming, Operating Systems, and Computer Networks.
- Developing full-stack applications with robust REST/gRPC API architectures.
- Integrating Vector Databases (Pinecone, Milvus) and securing AI engineering / RAG pipeline internships.

■ Career & Placements Strategy

- Completing an industry-oriented final-year project featuring an optimized generative AI workflow.
- Preparing for technical interviews covering system design, cloud MLOps, and core CS fundamentals.

■ Future-Proof Career Paths (Target 2030)

Core Focus: Model Integration Specialist | AI Application Engineer | Backend MLOps Engineer

Secondary Paths: Data Infrastructure Engineer | Cloud Solutions Architect | Full-Stack AI Developer

Long-Term Path: CSE Student → AI/Backend Intern → Model Integration Specialist → AI Systems Engineer → Principal AI Architect → CTO

■ Connect With Me

Email: harshithavenugopal3107@gmail.com

Location: Chennai, India

LINKEDIN PROFILE SUMMARY

Harshitha Venugopal

Headline: Computer Science Engineering Student | Aspiring Software Engineer
Email: harshithavenugopal3107@gmail.com | **Location:** Chennai, Tamil Nadu, India
LinkedIn URL: linkedin.com/in/harshitha-venugopal-638a8b406

Professional Summary

An analytical and detail-oriented Computer Science Engineering student with strong foundational programming knowledge backed by external multi-language certifications. Passionate about Data Structures & Algorithms (DSA) and building clean, efficient software. Proactively supplementing university academics with structured technical training to solve complex computing problems.

Technical Skills Matrix

Languages: Java, C++, C, Python

Core Focus: Data Structures & Algorithms (DSA), Problem Solving, Object-Oriented Programming (OOP)

Tools & Platforms: Git, GitHub, Visual Studio Code

Technical Experience & Training

CodeGalatta — Software Engineering & DSA Trainee (April 2026 – Present)

- Engaging in structured weekly training focused on foundational computer science paradigms and algorithmic design.
- Implementing and debugging abstract data structures including arrays, stacks, queues, and linked lists using Java and C++.
- Reviewing algorithmic runtime performance using Big O notation to write optimal time and space complexity code.

Licenses & Certifications

- Java Programming Certification — Online Verification
- C++ Programming Certification — Online Verification
- C Programming Certification — Online Verification
- Python Programming Certification — Online Verification

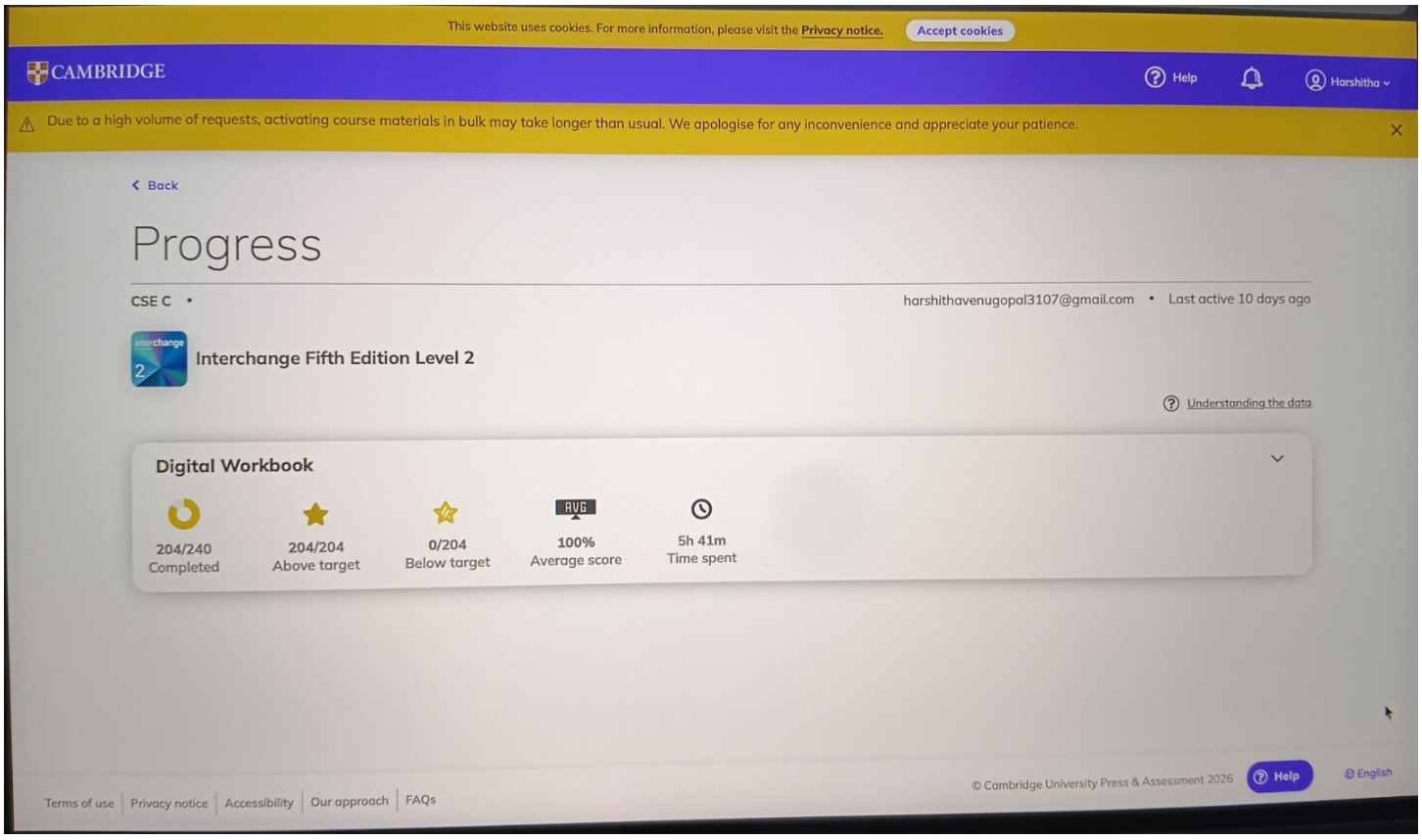
Education

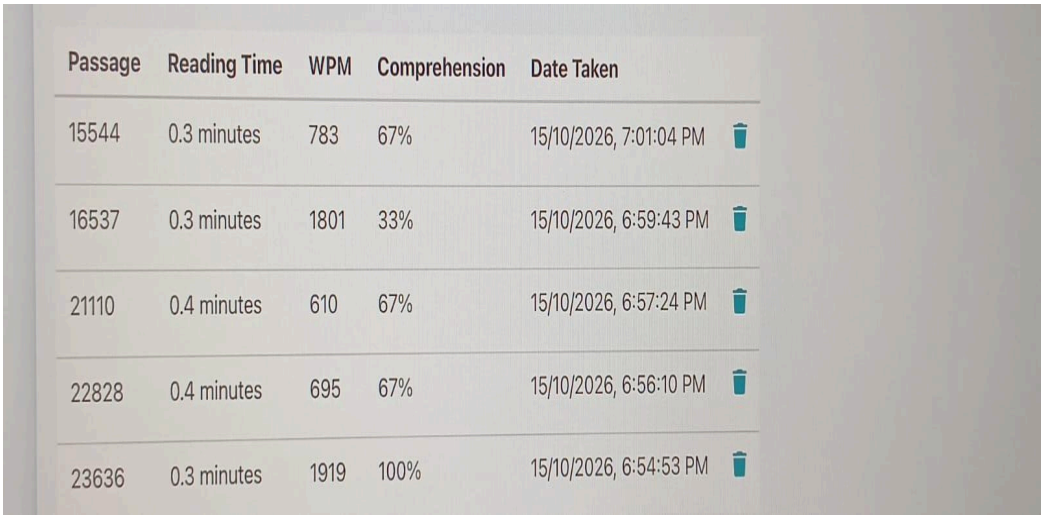
St. Joseph's College of Engineering (August 2026 – May 2030)
Bachelor of Engineering (B.E.) — Computer Science Engineering

Vidya Matric Higher Secondary School (Graduated: 2026)
Higher Secondary Course (Class 12)

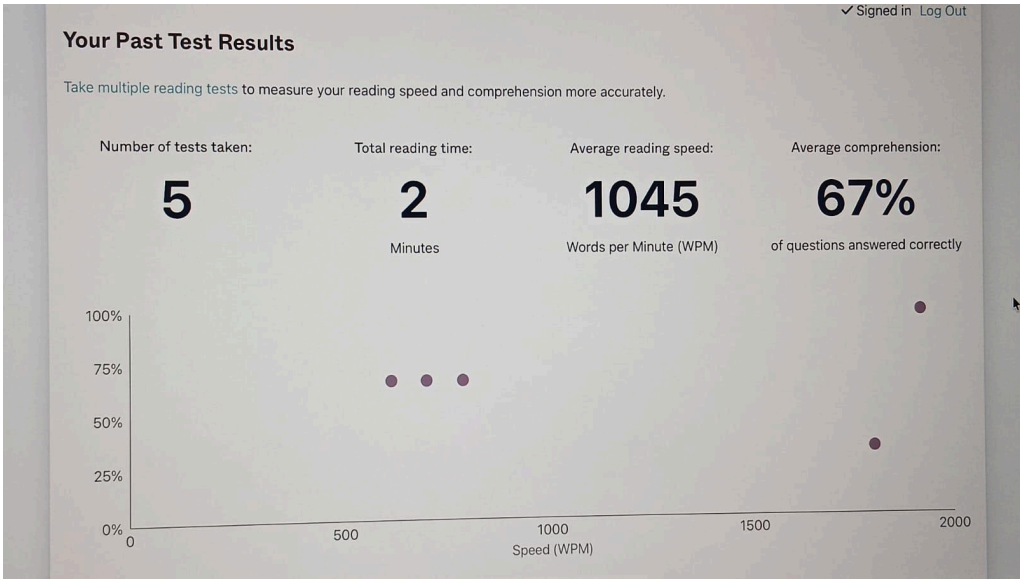
Vels Vidyasharam (Graduated: 2024)
All India Secondary School Examination (Class 10, CBSE)

CAMBRIDGE ONE – PROGRESS SCREENSHOT





Passage	Reading Time	WPM	Comprehension	Date Taken
15544	0.3 minutes	783	67%	15/10/2026, 7:01:04 PM
16537	0.3 minutes	1801	33%	15/10/2026, 6:59:43 PM
21110	0.4 minutes	610	67%	15/10/2026, 6:57:24 PM
22828	0.4 minutes	695	67%	15/10/2026, 6:56:10 PM
23636	0.3 minutes	1919	100%	15/10/2026, 6:54:53 PM



EN25C09: Professional English- I (LAB INTEGRATED) MODEL EXAM I –PRACTICE WORKSHEET PART - A (15 X 1 = 15 Marks)

Answer ALL questions. Each question carries 1 mark.

1. Fill in the blank with the correct form of the verb given in bracket:

The automated system _____ (monitor) the temperature continuously to ensure safe operation.

Answer: monitors

2. Fill in the blank with the correct past form of the verb given in bracket: The engineers

_____ (record) the vibration readings during yesterday's trial.

Answer: recorded

3. Complete the sentence with the appropriate comparative form of the adjective in bracket:

The newly developed composite material is significantly _____ (strong) than the conventional material while maintaining a lower weight.

Answer: stronger

4. Fill in the blank with an appropriate modal auxiliary verb: (must/can/might) Engineers

_____ follow safety regulations while operating heavy machinery.

Answer: must

5. Fill in the blank with a suitable modal auxiliary verb expressing capability: (can, may, might)

Modern sensors _____ detect even minor changes in temperature.

Answer: can

6. Combine the sentences into a single coherent sentence using a suitable relative pronoun:

(a) The laboratory developed a new testing device. (b) The device measures material strength with high precision.

Answer: The laboratory developed a new testing device which measures material strength with high precision.

7. Complete the sentence using the correct form of the verb in bracket (First Conditional – If Clause):

If the temperature _____ (rise) above the permissible limit, the cooling system will activate automatically.

Answer: rises

8. Complete the sentence using the correct form of the verb in bracket (Third Conditional – Past Unreal Conditional):

If the technician had inspected the circuit earlier, the equipment _____(not / fail) during operation.

Answer: would not have failed

9. Combine the following sentence using an infinitive of purpose (to + verb): The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

Answer: The mechanics installed dual damper springs to isolate chassis resonance.

10. Fill in the blank with the gerund form of the verb in bracket after the preposition: The research team reduced energy consumption by _____ (optimize) the motor control algorithm.

Answer: optimizing

11. Convert the following active voice into the Impersonal Passive Voice: The lab assistant

poured 50 mL of hydrochloric acid into the titration flask.

Answer: 50 mL of hydrochloric acid was poured into the titration flask.

12. Fill in the blank with the appropriate adverbial form derived from the adjective in bracket:

The robotic arm moves _____ (smooth) to ensure accurate positioning of the component.

Answer: smoothly

13. Rewrite the direct into indirect speech:

The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

Answer: The project supervisor asked the engineer if he had calibrated the pressure sensor before testing.

14. Identify the incorrect noun form and rewrite the sentence correctly:

The laboratory purchased several new equipments for conducting material testing.

Answer: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Fill in the blank using the correct irregular comparative form (better, worse, farther/further, less, more):

Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.

Answer: worse

PART - B (10 X 2 = 20 Marks)

Answer all questions. Each question carries 2 marks.

1. **Rewrite the following active operational and laboratory description in the Simple Past Tense:**

The engineer calibrates the pressure sensor and verifies its accuracy.

Answer: The engineer calibrated the pressure sensor and verified its accuracy.

2. Draft two comparative sentences comparing a laptop and a desktop computer in terms of portability and screen size.

Answer (Portability): A laptop is significantly more portable than a desktop computer because it is lightweight and features a built-in battery.

Answer (Screen Size): A desktop computer generally offers a much larger screen size than a laptop, providing an expansive display area for complex engineering tasks.

3. **Combine the technical statements into a single cohesive sentence using a defining or non defining relative clause:**

The engineer designed a new cooling system. It reduces the temperature of industrial machines.

Answer: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. **Complete the following conditional sentences logically:**

(a) If the battery voltage drops below the required level,
Answer : the backup generator will automatically switch on to maintain power continuity.

(b) If the technician had checked the connections earlier,
Answer : the system short circuit would have been completely avoided.

5. **Convert the following active sentences into the Impersonal Passive Voice suitable for formal engineering reports:**

The laboratory assistant heated the solution to 80°C before testing.

Answer: The solution was heated to 80°C before testing.

6. **Rewrite the following direct engineering query into Reported (Indirect) Speech:**

The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

Answer: The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.

7. **Identify and correct the grammatical errors involving mass/uncountable nouns in the following sentences:**

(a) *The engineer received many feedbacks on the new design.*
Answer: The engineer received a lot of feedback on the new design.

(b) *The students received several useful advices from their teacher.*
Answer: The students received several useful pieces of advice from their teacher.

8. **Rewrite the sentences using an Infinitive of Purpose or a Gerund construction:**
(a) *The technician wears safety gloves. He wants to protect his hands.*
Answer: The technician wears safety gloves to protect his hands.

(b) *The cooling fan operates continuously. It prevents the motor from overheating.*

Answer: The cooling fan operates continuously to prevent the motor from overheating.

9. **Fill in the blanks with the correct adverbial forms of the bracketed words:** a) The

machine operates **efficiently**(efficient) to reduce energy consumption.

(b) The technician checked the electrical connections **carefully** (careful) before starting the machine.

10. **Insert suitable modal auxiliary verbs to match the specified functional meaning:**

(a) The robotic arm **can**(capability) lift heavy components with precision.

(b) Visitors **must not**(strict prohibition) enter the testing area without authorization.

PART - C (5 X 13 = 65 Marks)

Answer all questions (Choose either 'a' or 'b' where options are provided). Each question carries 13 marks.

Question 26

· **26. a)** Write a set of eight clear, imperative safety instructions to promote safe and responsible use of the internet and protect users from cyber threats. *(13 Marks)*
- **Workspace / Planning Area:**

1. Create strong, unique passwords for every account using a mix of letters, numbers, and symbols.
2. Enable multi-factor authentication (MFA) on all sensitive professional and personal accounts.
3. Avoid clicking on suspicious links or downloading attachments from unknown email senders.
4. Update your operating system, web browsers, and security software regularly to patch vulnerabilities.
5. Verify the security of websites by checking for 'https://' in the URL before entering personal details.
6. Backup critical engineering data and personal files to a secure cloud or external hard drive frequently.
7. Restrict the amount of confidential or personal identifiable information (PII) you share on social media.
8. Disconnect from public Wi-Fi networks when handling private transactions, or use a secure VPN.

Question 27 (Choose either 27.a or 27.b)

27. a) Write a descriptive essay of 250 words about "A Traditional art form in your region and how modern initiatives can preserve its cultural significance." Discuss its historical origins, traditional practices, and ways to promote and sustain it among younger generations. *(13 Marks)*

OR

27. b) Write a structured travel plan and descriptive feature (250 words) on "An Educational Trip to a Place of Historical and Cultural Importance." Include travel arrangements, important landmarks, local food experiences, and observations of the region's cultural heritage and traditions. *(13 Marks)*

Workspace / Outline Area:
Selected Question: [27.b]
Introduction: An educational excursion to Mahabalipuram offers engineering and architecture students a unique chance to study ancient rock-cut cave temples and monolithic stone structures dating back to the 7th-century Pallava dynasty.
Body Paragraph 1: The trip begins with a private charter bus departure from the college campus at 07:00 AM, navigating the scenic East Coast Road. The core itinerary highlights architectural marvels including the iconic Shore Temple, the monumental relief carving of Arjuna's Penance, and the Pancha Rathas (Five Chariots). These landmarks expose students to brilliant ancient civil engineering and structural load-bearing principles executed directly on granite rock faces.
Body Paragraph 2: During the afternoon break, students will sample local coastal cuisine,

enjoying traditional South Indian thalis served on banana leaves alongside fresh, locally sourced seafood. Interacting with modern stone carvers keeping the UNESCO heritage craft alive provides rich insights into the region's living history. Students will actively record structural observations, sketch ancient stone joinery techniques, and interview local heritage caretakers.
Conclusion: This field study bridges the gap between historical craftsmanship and structural design principles, leaving students with a profound respect for India's enduring architectural heritage.

Question 28 (Choose either 28.a or 28.b)

28. a) Write a formal Letter of Request to the Librarian of your college seeking permission to extend library working hours during the semester examination period. Explain the academic benefits, the expected student participation, and the arrangements required to ensure proper use of library facilities. *(13 Marks)*

OR

28. b) Draft a formal Letter of Request to the Principal of your college seeking permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by a reputed institution. Mention the objectives of participation, the expected learning outcomes, and the arrangements required. *(13 Marks)*
Workspace / Format Outline:
From:

Student Coordinator, Department of Information Technology, St. Joseph's College of Engineering, OMR, Chennai - 119.

Date: 2 November 2026

To: The Principal, St. Joseph's College of Engineering, OMR, Chennai - 119.
Subject: Request for permission to attend a technical workshop on AI and Emerging Technologies

Respected Sir,
I am writing to request your kind permission for a group of five meritorious third-year students to participate in the upcoming National Workshop on Artificial Intelligence and Emerging Technologies, hosted by IIT Madras on November 15th and 16th.

The main objective of our participation is to gain practical exposure to machine learning algorithms and advanced neural networks directly from industry experts. Attending this event will allow us to represent our institution on a prestigious platform, develop innovative ideas for our final-year projects, and bring back valuable knowledge to share with our peers through a departmental seminar. The students have already arranged to cover their missed classwork and lab sessions during weekend hours.

We kindly request your approval and permission to grant us academic attendance for these two days.

Thank you.

Yours respectfully,
Student Coordinator

Question 29 (Choose either 29.a or 29.b)

29. a) Compose an analytical essay (250–300 words) comparing the advantages and disadvantages of Artificial Intelligence–Based Assessment versus Traditional Teacher-Based Assessment in engineering education. Evaluate assessment accuracy, feedback time, fairness, and the role of human judgement. *(13 Marks)*

OR

29. b) Write a compare-and-contrast essay of 250–300 words analysing the advantages and disadvantages of Living in Urban Areas versus Rural Areas. Discuss employment opportunities, cost of living, access to education and healthcare, and quality of life. *(13 Marks)*

· Workspace / Comparative Matrix & Outline:
Criteria 1 vs. Criteria 2: AI-Based Automation vs. Traditional Human Grading
Pros & Cons Breakdown: AI provides immediate, objective feedback but lacks qualitative empathy; Human grading offers deep, contextual feedback but takes significant time and can be prone to fatigue bias.

Essay:
The introduction of Artificial Intelligence (AI) into engineering education has sparked a major debate regarding whether automated evaluation systems can replace traditional, teacher-based grading. Evaluating these formats across key metrics like accuracy, turnaround time, fairness, and structural judgment highlights distinct advantages and drawbacks for both methods.

In terms of feedback time and evaluation speed, AI-based systems are unmatched. Code compilers and automated marking engines evaluate hundreds of student submissions instantly, allowing engineering students to correct mistakes in real time. Furthermore, AI ensures absolute consistency and fairness, removing any chance of personal bias or grader fatigue. However, automated tools struggle with nuanced assessment accuracy when evaluating open-ended engineering design concepts or creative problem-solving methods.

This is where traditional teacher-based assessment remains essential. A human professor brings vital professional judgment to the table, allowing them to evaluate the logic behind a partially incorrect formula or appreciate a unique, non-traditional approach to a complex problem. While human grading takes longer and can sometimes be inconsistent, it delivers holistic, supportive feedback that nurtures a student's technical growth. Ultimately, engineering education thrives best on a hybrid approach, using AI for prompt objective evaluations while relying on human faculty for deep mentorship and complex design assessments.

Question 30 (Choose either 30.a or 30.b)

30. a) Draft a personal email to your school friend sharing your experiences during the first semester of your engineering course. Describe your new friendships, classroom activities, adjustment to college life, and expectations for the upcoming semester examinations. *(13 Marks)*

OR

30. b) Draft a personal email to your school friend describing your experience attending the orientation programme for first-year engineering students at your college. Highlight the activities conducted, the introduction to academic facilities and college regulations, and how the programme helped you understand college life and prepare for your academic journey. *(13 Marks)*

· Workspace / Email Structure:
To: sneha.m@email.com
Subject Line: Incredible orientation week at college!
Dear Preethi,

I hope everything is great with you. I just finished my first official week at college and couldn't wait to tell you all about our freshers' orientation programme.

The orientation was incredibly well organized and ran over three action-packed days. The college held interactive ice-breaking activities that made it easy to meet my new classmates and overcome those initial first-day nerves. We were taken on guided tours of the campus, including the massive central workshops, advanced computer laboratories, and the three-story library.

The professors also held detailed briefings explaining the academic credit systems and strict campus regulations. This week really helped demystify college life and gave me a clear roadmap for the academic journey ahead. I now feel much more confident and genuinely excited to start our regular lectures next Monday.

Tell me all about your college opening as well. Write back when you get a chance!

Best wishes,
Harshitha

You Choose, We Do It
St. JOSEPH'S COLLEGE OF ENGINEERING
(An Autonomous Institution)
St. Joseph's Group of Institutions
OMR, CHENNAI - 119

Professional English - I (Lab Integrated)
Model – I – Practice Worksheet

Course Code: EN25CO9 | Subject: Professional English - I (Lab Integrated)

Exercise 1: Simple Present Tense (Subject-Verb Concord)

Fill in the blanks with the correct form of the verbs given in brackets:

1. Neither the project supervisor nor the lead programmers **recognize** (recognize / recognizes) the anomaly in the build script.
2. Each of the automated sorting arms **requires** (require / requires) recalibration after twelve hours of continuous duty.
3. Either the primary sensor or the auxiliary telemetry units **transmit** (transmit / transmits) packet data to the ground station.
4. The consortium of aerospace contractors **agrees** (agree / agrees) to adhere to the revised tolerance levels.
5. Physics and material science **form** (form / forms) the core theoretical foundation for nanoscale fabrication.

Exercise 2: Simple Past Tense (Trial Reporting)

Fill in the blanks with the simple past form of the given verbs:

1. Last month, the metallurgy division **conducted** (conduct) destructive fatigue testing on the titanium alloy shaft.
2. The field engineers **deployed** (deploy) thirty IoT water-monitoring nodes across the river basin.
3. During the pressure overload trial, the relief valve **opened** (open) at 150 psi as designed.
4. The research unit **published** (publish) their initial benchmark results on low-power RISC-V architectures.

5. In the 2025 dry run, the autonomous navigation algorithm **detected** (detect) all dynamic road obstacles successfully.

Exercise 3: Comparative Adjectives (Technical Metrics)

Complete each sentence with the appropriate comparative form of the adjective in brackets:

1. Gallium nitride (GaN) semiconductors operate at **higher** (high) power efficiency than standard silicon transistors.

2. Solid-state batteries deliver a **safer** (safe) operational threshold under rapid discharge cycles than traditional lithium cells.

3. The revised compiler generates code that executes **faster** (fast) than the legacy unoptimized routine.

4. Brushless DC motors are considerably **more durable** (durable) than brushed equivalents due to reduced mechanical wear.

5. The liquid cooling loop maintained a **lower** (low) core temperature than the dual fan-sink array.

Exercise 4: Modals of Obligation & Necessity

Fill in each blank with an appropriate modal auxiliary (must, should, or have to / has to):

1. Every operator **must** wear anti-static wrist straps before handling open CMOS motherboards.

2. Hazardous chemical waste **should** be neutralized prior to disposal into the industrial drain network.

3. Field technicians **have to** log their entry time into the high-voltage testing bay without fail.

4. To maintain ISO certification, calibration logs **must** be audited biannually by an external inspector.

5. The test rig **has to** undergo an emergency stop inspection before initiating the high-speed turbine run.

Exercise 5: Modals of Capability & Possibility

Fill in each blank with a suitable modal expressing capability or possibility (can, could, may, might):

1. With the upgraded DSP chip, the system **can** process 4K video streams in real time.

2. If thermal throttling occurs, the processing clock rate **might** drop by as much as 30%.

3. Advanced laser profilometry **can** measure surface roughness down to sub-micron tolerances.

4. Unshielded signal lines **may** pick up electromagnetic interference from nearby induction coils.

5. The modular chassis design **can** accommodate up to four secondary expansion cards.

Exercise 6: Relative Clauses

Combine each pair of sentences into a single coherent sentence using a suitable relative pronoun (who, which, that, whose):

1. (a) A civil engineer designed the suspension cable bridge. (b) She received the national infrastructure award.

   Answer: The civil engineer who designed the suspension cable bridge received the national infrastructure award.

2. (a) The spectrometer analyzes infrared emission spectra. (b) It is mounted inside the cleanroom vacuum chamber.

   Answer: The spectrometer, which is mounted inside the cleanroom vacuum chamber, analyzes infrared emission spectra.

3. (a) The technical specialist was reassigned to the robotics lab. (b) His patent on pneumatic grippers was approved.

   Answer: The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.

4. (a) This microcontroller features ultra-low power sleep states. (b) It is ideal for battery-operated remote sensors.

   Answer: This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.

5. (a) The field researchers surveyed the wind turbine farm. (b) They submitted their acoustic emission assessment yesterday.

   Answer: The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1 – Real Conditionals)

Complete the sentences with the correct form of the verbs in brackets:

1. If the bus voltage **exceeds** (exceed) 24V, the zener diode will divert the surge to ground.

2. The watchdog timer will reset the CPU if the main thread **fails** (fail) to ping it within 500 ms.

3. If the operator **engages** (engage) the safety latch, the hydraulic press will not actuate.

4. If temperature levels **rise** (rise) above 90°C, the cooling fans will switch to maximum RPM.

5. The system will trigger an auditory alarm if the toxic gas detector **registers** (register) more than 20 ppm.

Exercise 8: If-Clauses (Type 3 – Unreal Past / Troubleshooting)

Fill in the blanks with the past unreal conditional form (had + past participle / would have + past participle):

1. If the backup generator **had started** (start) immediately, the data centre servers would not have shut down.

2. The turbine bearing would not have seized if the maintenance crew **had lubricated** (lubricate) it as scheduled.

3. If the security team had patched the zero-day exploit earlier, the data breach **would have been prevented** (prevent).

4. The circuit breaker **would have tripped** (trip) safely if the overcurrent sensor had functioned properly.

5. If the telemetry link **had remained** (remain) stable, ground engineers would have corrected the satellite's orbital trajectory.

Exercise 9: Infinitives of Purpose

Combine the two ideas using an infinitive of purpose (to + verb):

1. (a) The mechanics installed dual damper springs. (b) They wanted to isolate chassis resonance.

   Answer: The mechanics installed dual damper springs to isolate chassis resonance.

2. (a) Technicians purge the fuel line with dry nitrogen gas. (b) They want to remove residual flammable vapors.

   Answer: Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.

3. (a) We introduced an optical encoder. (b) We intended to determine spindle velocity accurately.

Answer: We introduced an optical encoder to determine spindle velocity accurately.

4. (a) The developers wrote an automated load testing script. (b) Their goal was to simulate ten thousand concurrent connections.

Answer: The developers wrote an automated load testing script to simulate ten thousand concurrent connections.

5. (a) Gold plating is applied to terminal connectors. (b) Its purpose is to prevent atmospheric corrosion.

Answer: Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

Fill in the blanks with the gerund form of the bracketed verb after the preposition:

1. The aerodynamic team improved aircraft fuel economy by **redesigning** (redesign) the winglet profile.

2. Always disconnect the power supply before **replacing** (replace) defective surface-mount components.

3. The facility was praised for **reducing** (reduce) volatile organic chemical emissions by 40%.

4. Avoid **cleaning** (clean) precision optical lenses with abrasive industrial solvents.

5. The digital signal processor isolates target audio signals by **filtering** (filter) out background ambient noise.

Exercise 11: Impersonal Passive Voice

Convert the following active laboratory/industrial actions into the Impersonal Passive Voice:

1. The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Answer: 50 mL of hydrochloric acid was poured into the titration flask.

2. The automation unit tested the robotic arm under a 50 kg dynamic payload.

Answer: The robotic arm was tested under a 50 kg dynamic payload by the automation unit.

3. Field engineers calibrated the optical pyrometer prior to ignition.

Answer: The optical pyrometer was calibrated prior to ignition.

4. The technicians observed severe cavitation damage along the impeller blades.

Answer: Severe cavitation damage was observed along the impeller blades.

5. We store volatile solvents in flame-resistant containment cabinets.

Answer: Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

Fill in the blanks with the appropriate adverbial form derived from the adjective in brackets:

1. The stepper motor rotates **precisely** (precise) in 1.8-degree angular increments.

2. The telemetry ground station **periodically** (periodic) polls all deployed marine buoy sensors.

3. The thermal imaging camera registered the temperature drop **rapidly** (rapid) during quenching.

4. Structural joints must be welded **firmly** (firm) to withstand heavy cyclic vibrational stress.

5. The autonomous drone traversed the confined pipeline corridor **smoothly** (smooth).

Exercise 13: Reported Speech (Interrogative Structures)

Transform each direct technical query into indirect/reported speech:

1. The quality manager asked the technician, "Has the insulation resistance test been documented?"

Answer: The quality manager asked the technician whether the insulation resistance test had been documented.

2. The lead architect enquired, "Can the server cluster handle double the transaction throughput?"

Answer: The lead architect enquired if the server cluster could handle double the transaction throughput.

3. The field auditor asked the safety engineer, "Did you inspect the gas leakage relief lines?"

Answer: The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.

4. The client asked the team, "Will the firmware update fix the intermittent battery drain?"

Answer: The client asked the team if the firmware update would fix the intermittent battery drain.

5. The instructor asked the lab group, "Why did the multimeter show fluctuating voltage readings?"

Answer: The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

Identify the incorrect noun form and rewrite each sentence correctly:

1. The procurement officer purchased several new machineries and equipments for the workshop.

Answer: The procurement officer purchased several new pieces of machinery and equipment for the workshop.

2. The laboratory generated a huge amount of test datas during the wind tunnel trials.

Answer: The laboratory generated a huge amount of test data during the wind tunnel trials.

3. We need three more softwares licenses to complete the finite element analysis on the model.

Answer: We need three more software licenses to complete the finite element analysis on the model.

4. The field report provided numerous valuable informations regarding the soil composition.

Answer: The field report provided numerous valuable pieces of information regarding the soil composition.

5. All chemical apparatuses and glasswares must be washed thoroughly after the experiment.

Answer: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

Fill in the blanks using the correct irregular comparative form (better, worse, farther/further, less, more):

1. Because the heat sink had poor thermal paste contact, heat dissipation was **worse** (bad) than expected.

2. The redesigned microstrip antenna achieved a **better** (good) signal-to-noise ratio than the baseline patch.

3. The autonomous vehicle traveled 50 km **farther** (far) on a single charge after the weight reduction overhaul.

4. Switching to asynchronous task queues resulted in **less** (little) CPU idle time during peak batch processing.

5. We require **further** (further / father) metallurgical investigation into why the stress cracks formed along the weld seam.

Part B

Exercise 1: Tenses (Past Tense Conversion & Industrial Reporting)

Rewrite the following active operational and lab descriptions into the Simple Past Tense:

1. **The technician inspected the solar arrays and recorded the peak power conversion value.**
2. **The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.**
3. **Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.**
4. **The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.**
5. **The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.**

Exercise 2: Comparative Adjectives (Technical Metrics & Trade-Offs)

Draft technical comparative sentences based on the prompts provided:

1. **Cloud Storage vs. On-Premise Physical Servers**
    - **Initial Deployment Cost:** Cloud storage requires a significantly **lower** initial capital expenditure than on-premise physical servers, which demand substantial upfront infrastructure investments.
    - **Elastic Scalability:** Cloud storage offers far **more elastic** scalability than traditional on-premise hardware, allowing engineering teams to

adjust computational capacities instantaneously.

2. **Fiber-Optic Cables vs. Copper Twisted-Pair Cables**
    - **Data Transfer Bandwidth:** Fiber-optic cables deliver a vastly **higher** data transfer bandwidth than copper twisted-pair alternatives over long distances.
    - **Susceptibility to Electromagnetic Interference:** Fiber-optic networks are completely immune to electromagnetic interference, making them inherently **safer** and more reliable in high-voltage laboratory environments than copper wiring.
3. **Electric Vehicles (EVs) vs. Internal Combustion Engine (ICE) Vehicles**
    - **Tailpipe Emissions:** Electric vehicles are substantially **cleaner** regarding operational sustainability because they produce zero tailpipe emissions compared to ICE vehicles.
    - **Initial Purchase Price:** The initial purchase price of an electric vehicle remains noticeably **higher**than that of a conventional internal combustion engine car due to battery manufacturing expenses.
4. **Hydraulic Actuators vs. Electromechanical Actuators**
    - **Raw Force Output:** Hydraulic actuators yield a much **greater** raw force output than electromechanical units of comparable sizing constraints.
    - **Risk of Fluid Leakage:** Electromechanical actuators pose a drastically **lower** risk of fluid contamination, making them preferable for pristine laboratory cleanrooms.
5. **Relational (SQL) Databases vs. Document (NoSQL) Databases**
    - **Schema Rigidity:** Relational databases maintain a far **stricter** schema configuration than flexible NoSQL document stores.
    - **Horizontal Query Throughput:** Document databases scale horizontally with much **greater** ease, facilitating faster query throughput across vast distributed systems.

Exercise 3: Relative Clauses (Sentence Synthesis)

Combine each pair of technical statements into a single cohesive sentence using a defining or non-defining relative clause:

1. The algorithm analyzes microscopic cracks in turbine blades **which** were patented by the aeronautics team.
2. The lead chemical engineer, **who** noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator **where** high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant **whose** firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system utilizes an artificial neural network **that** predicts

transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses (Technical Conditionals & Troubleshooting)

Complete the following conditional statements logically and grammatically to depict realistic engineering scenarios:

1. (a) If the operating temperature exceeds $85^{\circ}C$, **the thermal overload protection relay will trip the main circuit breaker automatically.**
   (b) If the operator had pressed the emergency shutdown switch immediately, **the catastrophic pressure failure in the boiler system would have been completely averted.**
2. (a) If the supply voltage drops below the 200V threshold, **the digital under-voltage sensor initiates a safe shutdown cycle.**
   (b) If the network administrator had enabled multi-factor authentication, **the sophisticated ransomware attack on the mainframe server would not have succeeded.**
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, **the automated auxiliary ventilation fans will switch to maximum extraction capacity.**
   (b) If the mechanical team had checked the bearing lubrication schedule, **the main conveyor drive shaft would not have seized during production.**
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, **the onboard proximity logic module applies emergency regenerative braking.**
   (b) If the thermal sensor had registered the runaway reaction in time, **the chemical containment unit would have deployed the liquid nitrogen coolant.**
5. (a) If the software receives an unauthenticated API token, **the gateway firewall immediately terminates the network handshake protocol.**
   (b) If the civil inspectors had reviewed the core sample density tests, **they would have identified the foundation structural flaws before pouring the top concrete slab.**

Exercise 5: Impersonal Passive Voice

Convert the following active sentences into the objective Impersonal Passive Voice suitable for formal engineering reports:

1. **The crude extract was distilled at a constant temperature of $60^{\circ}C$.**
2. **The cross-section of the weld seam was polished with fine diamond paste.**

3. **Fifteen cast-iron pump casings were rejected due to internal porosities.**
4. **The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.**
5. **The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.**

Exercise 6: Reported Speech (Interrogative Technical Transformations)

Transform the following direct engineering queries into Reported (Indirect) Speech:

1. The safety supervisor asked the intern **whether he had verified the grounding connections on the generator.**
2. The plant director asked the site engineer **if the emergency effluent treatment plant had received clearance.**
3. The IT security lead enquired of the developer **why he had disabled the firewall rule on the database server.**
4. The client asked the project manager **if the bridge structure could withstand Category 4 hurricane winds.**
5. The research guide asked the postgraduate scholar **how many sample trials he had conducted under cryogenic conditions.**

Exercise 7: Countable vs. Uncountable Nouns

Identify and correct the grammatical errors involving mass/uncountable nouns in the following sentences:

1. (a) *Incorrect: software, machineries* → **We need to install several new software licenses and pieces of machinery in the CAD laboratory.**
   (b) *Incorrect: informations* → **The team collected plenty of useful data and information from the field test.**
2. (a) *Incorrect: apparatuses, equipments* → **The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.**
   (b) *Incorrect: infrastructures, powers* → **High-speed train operations require advanced transport infrastructure and dedicated electrical power.**
3. (a) *Incorrect: datas, pollutions* → **The environmental sensor gathers extensive atmospheric data and pollution indices every minute.**
   (b) *Incorrect: sands, cements* → **The construction crew unloaded tons of sand, cement, and steel debris at the worksite.**
4. (a) *Incorrect: hardwares, wirings, legislations* → **All electrical hardware and wiring must comply with local safety legislation.**
   (b) *Incorrect: knowledges* → **The data scientist discovered several interesting insights and pieces of knowledge after running the machine learning clustering.**

5. (a) *Incorrect: staffs, glasswares, wastes* → **Laboratory staff must clear away all glassware and acidic waste before leaving the premises.**
   (b) *Incorrect: fundings, researches* → **The project failed due to a lack of proper funding and technical research.**

Exercise 8: Infinitives of Purpose & Gerund Constructions

Rewrite the following pairs of sentences using either an Infinitive of Purpose (to + verb) or a Gerund construction after a preposition (by / for + -ing):

1. (a) **A pressure relief valve is installed to vent excess boiler steam.**
   (b) **The automated script runs every midnight for archiving legacy server logs.**
2. (a) **Automotive engineers installed an intercooler to reduce the compressed air intake temperature.**
   (b) **The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.**
3. (a) **High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.**
   (b) **The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.**
4. (a) **Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.**
   (b) **The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.**
5. (a) **The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.**
   (b) **The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.**

Exercise 9: Adverbs of Manner & Operational Precision

Fill in the blanks with the appropriate adverbial forms derived from the words provided in brackets:

1. (a) The drone adjusted its flight path **automatically** to avoid overhead power cables.
   (b) The software patch was integrated **flawlessly** into the production server.
2. (a) The CNC cutting head traced the complex automotive profile **accurately**.
   (b) The turbine governor responded **instantly** to the unexpected grid frequency drop.
3. (a) The hazardous gas extraction vents operate **continuously** throughout the electroplating process.
   (b) Technicians handled the radioactive isotope capsule **cautiously** using robotic arm manipulators.

4. (a) The optical pyrometer recorded the furnace temperature **consistently** across three production shifts.
   (b) The structural frame buckled **catastrophically** under the simulated earthquake test load.
5. (a) The firmware routine checks for incoming serial packets **periodically** every hundred microseconds.
   (b) The titanium alloy workpiece was machined **smoothly** to a mirror finish.

Exercise 10: Modals of Capability, Obligation, and Prohibition

Complete the sentences by inserting an appropriate modal auxiliary verb that conveys the specific operational meaning specified in brackets:

1. (a) The microcontroller **can** drive up to six stepper motors simultaneously.
   (b) Laboratory staff **must not** handle corrosive reagents without full face shields.
2. (a) Before entering the cleanroom chamber, all personnel **must** put on antistatic footwear.
   (b) Given the high ambient humidity, water droplet condensation **could** form inside the optical housing.
3. (a) Unauthorized maintenance personnel **shall not** override the high-voltage interlock switch.
   (b) The upgraded wireless transmitter **could** broadcast telemetry signals up to forty kilometers.
4. (a) Quality inspection teams **should** recalibrate measuring calipers every ninety days.
   (b) The digital ammeter is reading zero amperes; the auxiliary fuse **must** be blown.
5. (a) Industrial robots **can** perform repetitive spot welds with sub-millimeter precision.
   (b) Visitors **must not** enter the laser testing enclosure while the amber warning beacon is active.

Topic 1: Writing Imperative Safety Instructions (CO1 | BL3)

Target: 8 clear, numbered, imperative instructions using modals/action verbs.

Question 1 : Write a set of eight clear, imperative safety instructions to be strictly displayed and followed inside a Chemistry & Materials Testing Laboratory to avoid chemical burns, inhalation hazards, and equipment contamination.

1. **Wear** certified chemical splash goggles, a full-length lab coat, and nitrile gloves at all times inside the testing bay.
2. **Handle** all volatile, corrosive, or toxic solvents strictly inside the operational fume hood with the sash pulled down to the marked safety level.

3. **Never pipet** chemical solutions or liquid reagents by mouth; always utilize a mechanical pipet filler or automated electronic dispenser.
4. **Pour** concentrated acids slowly into water when preparing dilutions; never reverse this sequence to prevent highly exothermic splattering.
5. **Wash** all contaminated glassware thoroughly with deionized water and designated laboratory detergents before placing them in the hot-air drying cabinet.
6. **Dispose** of hazardous chemical byproducts, organic solvents, and heavy metal solutions inside their respective color-coded waste containers.
7. **Locate** and verify the pressure gauges on the emergency eye-wash station and chemical safety shower before initiating any material breakdown tests.
8. **Report** all chemical spills, glass breakages, or skin contact accidents immediately to the lab supervisor, regardless of how minor they appear.

Topic 2: Free Writing – Festivals & Cultural Celebrations (CO2 | BL4)

Target: Descriptive essay of 250 words integrating community customs, festive meals, and
sustainable practices.

Question 1 : Write a descriptive essay of 250 words about "A Traditional Harvest Festival in Your Region and How Modern Eco-Friendly Practices Can Preserve Its Core
Spirit." Discuss community customs, traditional festive meals, and sustainable celebration methods.

The traditional harvest festival of Pongal serves as a cultural cornerstone in Tamil Nadu, celebrating agrarian success and expressing gratitude to nature. Historically, this four-day celebration features communal cooking where newly harvested rice is boiled in traditional clay pots decorated with fresh turmeric plants. As the boiling milk foams over the rim, families gather to chant "Pongalo Pongal," welcoming prosperity. Festive meals are traditionally served on organic banana leaves, highlighting a menu of sweet jaggery rice, freshly harvested vegetables, sugarcane, and native lentils, which deepens the community's bond with agrarian cycles.

In the modern era, balancing these deep-rooted customs with contemporary environmental sustainability has driven valuable eco-friendly initiatives. Local community centers now organize centralized, smoke-free events to replace the traditional burning of household waste during the Bhogi phase, drastically lowering seasonal air pollution. Rural and suburban youth groups actively promote the use of organic rice flour instead of synthetic chemical powders to draw traditional kolam floor patterns, protecting local soil and insect biomes. Additionally, community feasts enforce a strict zero-plastic policy by using compostable areca leaf plates and terracotta cups for serving food and water.

These sustainable approaches show that modern environmental awareness can protect the core spirit of heritage, ensuring that the celebration of nature does not come at the expense of our natural environment.

Topic 3: Free Writing – Food & Technical Travel Plans (CO2 | BL4)

Target: Structured travel narrative (250 words) combining route logistics, local dining experiences, and technical landscape observations.

Question 1 : Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to a Major Hydroelectric Dam." Include route details, logistics, local cultural food experiences, and observations of the plant's operational landscape.

- **Route Details & Logistics:** The field visit covers a short coastal route along the East Coast Road to the Nemmeli 150 MLD Desalination Facility, using an air-conditioned college bus for student transportation.
- **Local Culinary Experiences:** The team will take a lunch break at a coastal fishing village restaurant, enjoying traditional banana-leaf meals with shallow-fried king fish, crab masala, and spiced buttermilk.
- **Technical Landscape Observations:** The industrial layout features a massive marine intake pumping station that draws raw seawater from 1 km offshore. Students will track the water flow through the pre-treatment stages, including dissolved air flotation (DAF) units and dual-media pressure filters. The core technical highlight is the high-pressure pump room, where seawater is pressurized up to 70 bar to push through banks of semi-permeable polyamide reverse osmosis membranes. Students will also inspect the energy recovery devices (ERDs) that recycle pressure from the concentrated brine stream, before finishing at the post-treatment remineralization columns and the offshore brine discharge diffuser pipelines.

Topic 4: Formal Letters of Request (CO3 | BL4)

Target: Formal business letter layout with sender/receiver address, clear subject line, technical
rationale, specific requirements, and courteous sign-off.

Question 1 : Draft a formal Letter of Request to the Head of the Computer Science Department requesting permission and lab access after college hours for your project team to prepare for a National AI Hackathon. Outline project objectives and required software resources.

From:

The Project Team Leader, Third-Year B.E CSE,

St. Joseph's College of Engineering,

OMR, Chennai - 119.

Date: 2 November 2026

To:

The Head of the Department,

Department of Computer Science and Engineering,

St. Joseph's College of Engineering,

OMR, Chennai - 119.

Subject: Request for permission and laboratory access after college hours for National AI Hackathon preparation

Respected Sir,

I am writing to request special permission for our four-member project team to utilize the Advanced Analytics Laboratory after standard college hours for the next two weeks, from 3 November to 17 November 2026, until 08:30 PM daily.

Our team has been shortlisted for the final round of the National AI Hackathon, which requires us to build and deploy a deep learning model for real-time computer vision analysis. To train our complex neural networks effectively, we need access to the high-performance desktop computers in Lab 4, which are equipped with dedicated NVIDIA RTX GPUs and pre-configured Linux environments. Our personal laptops lack the necessary computing power, leading to thermal throttling and prolonged training bottlenecks. Accessing the lab after hours will allow us to run extensive model training sessions without interrupting regular daytime classes.

We will ensure all safety protocols are strictly followed, and a student roster will be maintained to guarantee the lab is locked and all systems are powered down correctly each night. We hope you will support our academic endeavors and grant us this permission.

Thank you.

Yours respectfully,

[Signature]

Project Team Leader

Topic 5: Compare & Contrast Descriptive Essays (CO4 | BL5)

Target: Analytical essay (250–300 words) using comparative transitions (whereas, conversely,
in contrast, on the other hand) to evaluate technical trade-offs.

Question 1 : Write a comprehensive Compare and Contrast essay (250–300 words) analyzing the Positives and Negatives of Electric Vehicles (EVs) versus Conventional Internal Combustion Engine (ICE) Vehicles. Address battery life, manufacturing footprint, operating costs, and charging network feasibility.

Choosing between Electric Vehicles (EVs) and conventional Internal Combustion Engine (ICE) vehicles involves balancing key trade-offs across operational costs, infrastructure readiness, and long-term environmental impacts. While both transport models serve the same core purpose, their underlying engineering methods create distinct user experiences.

When evaluating operating costs, electric vehicles offer a significant advantage over conventional models; conversely, fuel costs for ICE vehicles remain tied to unpredictable global oil markets. EVs feature much simpler drivetrains with fewer moving parts, which lowers long-term maintenance requirements. In contrast, conventional internal combustion engines require regular oil changes, filter replacements, and complex transmission upkeep. However, analyzing battery life and manufacturing footprints reveals a different perspective. Producing high-capacity lithium-ion batteries for EVs requires energy-intensive mining for raw materials like cobalt and lithium, creating a larger initial carbon footprint than manufacturing standard ICE cars. Furthermore, replacing an aged EV battery pack after a decade introduces steep financial costs for consumers.

Infrastructure feasibility is another critical point of comparison. ICE vehicles benefit from a mature, worldwide network of fueling stations that allow drivers to fill their tanks in minutes. On the other hand, the charging network for electric vehicles is still developing, often leading to range anxiety during long-distance travel. Even with modern DC fast-charging technology, an EV requires significantly longer to charge compared to the quick refueling time of an ICE vehicle. In short, while electric vehicles offer an exceptionally clean, low-emission choice for urban commuters, conventional ICE vehicles remain highly practical for long-distance travel where charging infrastructure is limited.

Topic 6: Personal Email Writing (CO5 | BL6)

Target: Personal email (informal yet polite and articulate tone) conveying reflections, academic

transitions, and structured study plans.
Question 1 : Draft a Personal Email to your high school mentor or teacher sharing your first-semester experiences at engineering college. Highlight the transition to independent campus life, your initial laboratory projects, and how you are preparing for your upcoming end-semester examinations.

To: ram123@gmail.com

Subject: Reflections on my first semester at St. Joseph's Engineering College

Dear Sir,

I hope this email finds you in good health. As I approach the end of my first semester here at St. Joseph's College of Engineering, I wanted to share an update on my transition into college life and express my sincere gratitude for your guidance.

Moving away from home and adapting to independent campus life was challenging for the first few weeks, but I have established a productive daily routine. The academic environment here is demanding, yet highly rewarding. I am enjoying our first practical laboratory projects, particularly in the engineering graphics and basic programming labs, where we apply classroom concepts to solve real-world problems.Right now, the entire campus is preparing for the upcoming end-semester examinations. I have put together a structured revision plan to review our advanced calculus and engineering physics modules, and I am making frequent use of the college reference library. Thank you once again for pushing me to build strong academic fundamentals during high school; they have been incredibly helpful during this transition. I would love to hear how the new batch of students is doing at school when you have a moment.

Warm regards,

Harshitha V

First Year, B.E.CSE

Topic 7: Travel Planning & Budgetary Itineraries (CO5 | BL6)

Target: Organized travel schedule and budget breakdown (250 words) with day-by-day technical
activities and safety measures.
Question 1 : Write an engaging, well-organized Travel Plan and Budgetary Itinerary (250 words) for a 4-day student technical tour to an Aerospace and Satellite Launch

Station, detailing transport logistics, budget calculations, planned technical visits, and safety protocols.

- **Itinerary Schedule & Route Details:**
    - **Day 1: Depart from the college campus at 06:00 AM via a private luxury charter coach, traveling along the national highway to Nellore. Check into the guest house, followed by an afternoon safety briefing and an introductory lecture on orbital mechanics.**
    - **Day 2: Arrive at the Space Launch Center at 08:30 AM. Spend the morning exploring the Rocket Assembly Building and the solid propellant storage facilities. After lunch, visit the main launch pad control bunker and tracking radar array towers.**
    - **Day 3: Attend an expert technical lecture on satellite telemetry systems at the space museum auditorium. Spend the afternoon observing decommissioned rocket engines and payload fairing components.**
    - **Day 4: Participate in a final Q&A session with space center scientists. Depart the facility at 01:00 PM and return to the college campus by late evening.**
- **Budget Calculations (Per Student Estimates):**
    - ***Transport Logistics (Charter Coach):* ₹1,800**
    - ***Industrial Facility Entry & Processing Fees:* ₹400**
    - ***Guest House Lodging (3 Nights):* ₹1,500**
    - ***Catering Services (All Meals):* ₹1,200**
    - ***Total Estimated Expenditure:* ₹4,900 per student**
- **Safety Measures & Planned Technical Visits: All students must carry their official institutional ID badges and valid government-issued photo identification to pass multiple security checkpoints. Inside the facility, students must wear safety helmets, stay within designated yellow pedestrian pathways, and strictly avoid taking photos near active launch bays or tracking dishes.**